

## Literature Review

# Speech-Driven Drafting of First Information Reports for the Tamil Nadu Police:

*A Survey of Multilingual Speech Recognition, Legal Information Extraction, and Statute Identification*

Madhav K (23BAI1088) · Nischay Kuchibotla (23BAI1245) · Rohit A. S.  
(23BAI1416)

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### 1. Introduction and scope of the review

Drafting a First Information Report from a spoken complaint is not a single research problem but a chain of them. A usable system has to hear a complaint that is delivered mostly in Tamil, often mixed with English and rarely in a tidy register; recover from it the specific particulars that a statutory form demands; decide which offences the account discloses and under which sections of the new penal code; and finally produce a document that an officer can check line by line. Each of these steps corresponds to an active area of research, yet the areas have grown up largely apart from one another, and almost none of the published work has been evaluated on the conditions that a police station actually presents.

This review reads the relevant literature with one question in mind: how far do existing methods carry us toward a bilingual, explainable, officer-supervised FIR-drafting system for the Tamil Nadu format, and where do they stop? The discussion is organised by capability rather than by chronology. Section 2 examines speech recognition for Tamil and for code-switched speech. Section 3 turns to information extraction and named-entity recognition, with an emphasis on Indian-language and legal text. Section 4 surveys statute identification and the wider problem of legal judgment prediction. Section 5 considers the practical complication introduced by the 2023 recodification of Indian

criminal law, and the retrieval methods that help address it. Section 6 looks at faithful and controllable generation, which is what makes automatic drafting safe enough to attempt. Section 7 places the whole effort within the literature on human-in-the-loop decision support and data protection. Section 8 draws the threads together and states the gap the project is designed to fill.

## **2. Speech recognition for Tamil and code-switched speech**

Modern speech recognition rests on representations learned from large amounts of unlabelled or weakly-labelled audio. wav2vec 2.0 (Baevski et al., 2020) showed that a model pre-trained to predict masked latent speech units could be fine-tuned to competitive word-error rates with comparatively little transcribed data, and its cross-lingual successor XLS-R (Conneau et al., 2021) extended the idea to dozens of languages at once. Whisper (Radford et al., 2023) took a different route, training on a very large and diverse set of weakly-supervised audio-text pairs, and in doing so acquired a degree of robustness to accent and background noise that earlier systems lacked. For Tamil specifically, Whisper offers reasonable zero-shot transcription, but its accuracy falls on spontaneous and dialectal speech, which is precisely the kind a complainant produces.

The most useful Indian-language resources come from the AI4Bharat effort. The IndicSUPERB benchmark (Javed et al., 2023) and the Kathbath corpus provide read speech with transcriptions across twelve languages, including a substantial Tamil portion, and the Shrutilipi mining pipeline (Bhogale et al., 2023) adds a much larger, if noisier, body of aligned audio and text. Models such as IndicWhisper and IndicConformer, trained or adapted on these data, improve Tamil recognition appreciably over generic baselines. Standard evaluation is carried out on FLEURS (Conneau et al., 2023) and Mozilla Common Voice (Ardila et al., 2020). The recurring difficulty with all of these resources, for our purposes, is that they are dominated by read or broadcast speech; a citizen narrating a theft or a family dispute speaks quite differently, with hesitations, restarts, and frequent switches into English.

Code-switching is the sharper problem. Sitaram et al. (2019) survey the field and note that intra-sentential mixing breaks the assumptions of monolingual acoustic and language models, and that annotated code-switched speech is scarce for almost every language

pair. Tamil-English switching is common in exactly the tokens an FIR cares about, such as names, place names, vehicle registration numbers, and monetary amounts, so errors there are costly out of proportion to their frequency. A related and comparatively neglected requirement is post-processing: punctuation restoration and inverse text normalisation, which turn a raw transcript into something with usable dates, times, and figures. These steps are treated as an afterthought in most ASR papers, yet for form-filling they are not optional. In short, the acoustic modelling is largely solved for clean Tamil, but the register, the mixing, and the normalisation that our setting demands are not well served by the existing benchmarks.

### **3. Information extraction and named-entity recognition**

Transformer encoders underpin most current work on extracting structured information from text. Multilingual models such as mBERT and XLM-R (Conneau et al., 2020) transfer reasonably to Indian languages, and purpose-built encoders narrow the gap further: IndicBERT, released with the IndicNLP Suite (Kakwani et al., 2020), and MuRIL (Khanuja et al., 2021), which was trained on Indian-language text together with transliterated pairs, both give strong results on token-level tasks in Tamil and its neighbours. For the legal domain, LEGAL-BERT (Chalkidis et al., 2020) demonstrated that in-domain pre-training matters for legal text, a finding later echoed for Indian law.

Named-entity recognition tuned to Indian legal documents is directly relevant. Kalamkar et al. (2022), working within the OpenNyAI initiative, released a corpus annotated with fourteen legal entity types over court judgments, which supports high-quality recognition of courts, statutes, provisions, and parties. This is a valuable warm-start, but the mismatch with our task is real: the corpus is drawn from judgments written by judges, whose vocabulary and structure have little in common with a complainant's spoken account, and its entity schema is not the set of fields the IF-1 form requires. A separate line of work replaces bespoke taggers with large language models prompted to emit structured output under a schema, which is attractive because it can produce a filled record together with per-field confidence and pointers back to the source text. What the literature does not yet offer is an extraction target aligned to a specific statutory form, or

a treatment of uncertainty and provenance as first-class outputs rather than diagnostics. Both are central to a system whose output an officer must be able to trust and trace.

#### **4. Statute identification and legal judgment prediction**

Predicting the offences or statutes implicated by a set of facts has been studied most extensively on Chinese datasets. Luo et al. (2017) introduced an attention mechanism that reads the fact description in light of candidate charges; Zhong et al. (2018) modelled the dependencies among subtasks such as charges, applicable articles, and terms of imprisonment with their TopJudge framework; and the CAIL2018 dataset (Xiao et al., 2018) put the problem on a large-scale footing. These systems are accurate but opaque, and they operate on a legal system and language far removed from ours.

For Indian law, Malik et al. (2021) released the Indian Legal Documents Corpus and framed court-judgment prediction with explanation, an early recognition that a bare label is of little use to a legal reader without a reason. Statute identification in the narrower sense is addressed directly by the ILSI dataset and the LeSICiN model (Paul et al., 2022), which casts the task as multi-label classification and represents statutes and documents jointly in a heterogeneous graph so that citation structure informs prediction. Domain-adapted encoders, released as InLegalBERT and InCaseLawBERT (Paul et al., 2023), raise performance on such classification further. More recently, LegalBench (Guha et al., 2023) has begun to probe whether large language models can perform genuine legal reasoning rather than surface pattern-matching, with mixed results.

Three limitations of this body of work bear directly on the present project. First, the training signal is IPC-era: ILSI is labelled with sections of the Indian Penal Code, which the Bharatiya Nyaya Sanhita has since replaced, so the labels cannot be used operationally without translation. Second, the inputs are formal case facts distilled by lawyers and judges, not the colloquial spoken complaints our system receives, which raises an unresolved question of domain transfer. Third, the explanations these models provide, where they provide any, are typically attention weights or extracted rationales rather than a check of whether each statutory ingredient of an offence is actually present. A system meant to help an officer justify a charge needs the latter, and the literature offers little precedent for it.

## **5. Statutory change and access to legal knowledge**

A complication specific to the Indian setting, and very recent, is that the substantive and procedural criminal law changed in 2024. The Bharatiya Nyaya Sanhita, 2023 and the Bharatiya Nagarik Suraksha Sanhita, 2023 replaced the Indian Penal Code and the Code of Criminal Procedure with effect from July 2024, renumbering and reorganising the offences that every new FIR must cite. Almost no peer-reviewed natural-language-processing work yet targets the new code, which means that the corpora the community relies on, ILSI included, are now one legal generation out of date. The resources available for translating between the two are government correspondence tables and a growing set of practitioner compilations; in preparing this project we found that these agree on the common offences but diverge on sub-clauses and occasionally contradict one another, which argues for an explicit, versioned mapping that records confidence and defers uncertain cases to a human rather than a single lookup.

Reasoning over statutes also calls for reliable access to their text. Retrieval-augmented generation (Lewis et al., 2020) provides the standard mechanism: relevant passages are retrieved and supplied to a model so that its output is anchored to an authoritative source rather than to parametric memory. Dense multilingual retrievers such as BGE-M3 (Chen et al., 2024) make this practical across Tamil and English. The relevance of retrieval here is not only accuracy but verifiability, because an officer can be shown the exact statutory language against which a suggested section was checked.

## **6. Faithful and controllable generation**

Automatic drafting is only defensible if the generated text says nothing the evidence does not support. The tendency of neural generators to produce fluent but unsupported content is well documented: Maynez et al. (2020) showed that abstractive summarisers frequently introduce facts absent from the source, and the survey by Ji et al. (2023) catalogues the forms this hallucination takes across generation tasks. In a legal document a single fabricated detail is not a stylistic blemish but a fault, which raises the bar well above what summarisation research usually demands.

Three techniques, used together, make the risk manageable. Retrieval-augmentation keeps the model close to its sources. Constrained or guided decoding, formalised for example by Willard and Louf (2023), forces output to satisfy a schema, so an extractor can be made to emit only valid records and a narrative can be held to a controlled form. A separate verification pass, typically framed as natural-language inference, then checks that each generated statement is entailed by the cited evidence and rejects those that are not. None of these is individually novel, but their combination for form-filling, where the acceptable hallucination rate is effectively zero, has not been studied closely, and it is where the present work concentrates its attention. Making all of this run locally is feasible because parameter-efficient fine-tuning, through LoRA (Hu et al., 2022) and its quantised variant QLoRA (Dettmers et al., 2023), together with 4-bit inference, brings seven- and eight-billion-parameter models within reach of a single consumer GPU.

## **7. Decision support, accountability, and data protection**

The design choice that most distinguishes a responsible system in this domain is the decision to assist rather than to decide. A drafting tool that files reports or fixes charges on its own would concentrate consequential authority in an opaque model; one that proposes, explains, and records, while leaving every output editable and every action logged, keeps the officer accountable as the author of record. This framing is consistent with the broader literature on human oversight of automated decisions, which stresses contestability and the availability of reasons. It also has a statutory dimension in India: the Digital Personal Data Protection Act, 2023 imposes obligations of purpose limitation, data minimisation, and security that a system handling complainant information must satisfy, and which favour on-premise processing and strict retention limits.

Against this backdrop, the systems currently deployed are notably thin. The Crime and Criminal Tracking Network and Systems provides electronic entry of the IF-1 form, and some states expose citizen e-FIR portals, but these are data-entry interfaces without speech input, extraction, or any assistance with sections. The gap between what is deployed and what the research literature makes possible is therefore wide, and it is a gap in assistive, accountable tooling rather than in autonomous automation.

## 8. Synthesis and research gaps

Read together, the literature supplies strong components and no assembly. Speech recognition for Tamil is capable but is tuned to clean, read speech and stumbles on the code-switched, colloquial register of a real complaint. Encoders for Indian and legal text extract entities well, but from judgments rather than complaints, and against schemas that are not the FIR form. Statute identification is a mature task with good benchmarks, yet its data are labelled under the repealed Penal Code, its inputs are formal, and its explanations are shallow. Controllable and faithful generation offers the tools to draft safely, but has not been pushed to the near-zero-error setting that a legal form requires. And the systems actually in use do none of this.

The project this review supports occupies the space these gaps define. It targets spoken, bilingual, colloquial complaints rather than written case facts; it treats extraction as population of the specific IF-1 schema, with confidence and provenance attached to every value; it bridges IPC-era statute-identification data to the Bharatiya Nyaya Sanhita through an explicit, verified mapping; it replaces shallow explanation with an element-wise check of the ingredients of each offence; and it keeps the officer in the loop as the author of record, running locally so that it remains usable where connectivity does not reach. No prior work, to the best of this review, combines these properties, and it is that combination, rather than any single component, that the project contributes.

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